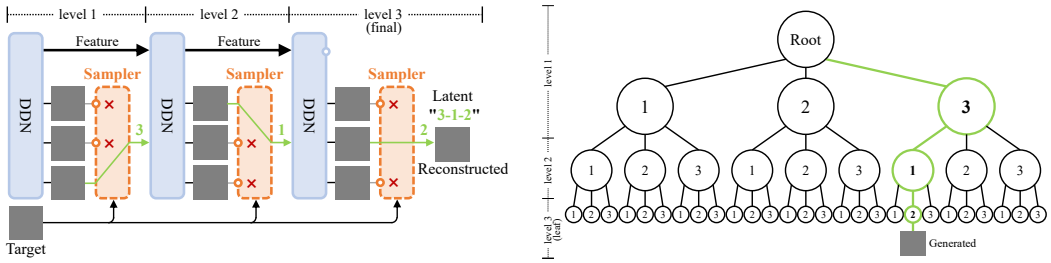
arXiv:2401.00036v3 [cs.CV] 16 Apr 2025

# DISCRETE DISTRIBUTION NETWORKS

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## ABSTRACT

We introduce a novel generative model, the Discrete Distribution Networks (DDN), that approximates data distribution using hierarchical discrete distributions. We posit that since the features within a network inherently capture distributional information, enabling the network to generate multiple samples simultaneously, rather than a single output, may offer an effective way to represent distributions. Therefore, DDN fits the target distribution, including continuous ones, by generating multiple discrete sample points. To capture finer details of the target data, DDN selects the output that is closest to the Ground Truth (GT) from the coarse results generated in the first layer. This selected output is then fed back into the network as a condition for the second layer, thereby generating new outputs more similar to the GT. As the number of DDN layers increases, the representational space of the outputs expands exponentially, and the generated samples become increasingly similar to the GT. This hierarchical output pattern of discrete distributions endows DDN with unique properties: more general zero-shot conditional generation and 1D latent representation. We demonstrate the efficacy of DDN and its intriguing properties through experiments on CIFAR-10 and FFHQ. The code is available at https://discrete-distribution-networks.github.io/




(a) Image Reconstruction through DDN

(b) Tree Structure of DDN's Latent

Figure 1: (a) Illustrates the process of image reconstruction and latent acquisition in DDN. Each layer of DDN outputs $K$ distinct images to approximate the distribution $P(X)$. The sampler then selects the image most similar to the target from these and feeds it into the next DDN layer. As the number of layers increases, the generated images become increasingly similar to the target. For generation tasks, the sampler is simply replaced with a random choice operation. (b) Depicts the tree-structured representation space of DDN's latent variables. Each sample can be mapped to a leaf node on this tree.

## 1 INTRODUCTION

With the advent of ChatGPT Brown et al. (2020) and DDPM Ho et al. (2020), deep generative models have become increasingly popular and significant in everyday life. However, modeling the complex and diverse high-dimensional data distributions is challenging. Previous methods Kingma & Welling (2014); Radford et al. (2016); Kingma & Dhariwal (2018); Goyal et al. (2021); Song et al. (2021); Shocher et al. (2024); Graves et al. (2023) have each demonstrated their unique strengths and characteristics in modeling these distributions. In this work, we propose a novel approach to model

samples simultaneously. Therefore, we adopt a strategy from autoregressive models Van Den Oord et al. (2016) and partition this large space into a hierarchical conditional probability model. Each layer of this model needs only a small number of outputs. We then select one of these outputs as the output for that layer and use it as conditional input to the next layer. As a result, the output of the next layer will be more closely related to the selected conditional sample. If the number of layers is $L$ and the number of outputs per layer is $K$, then the output space of the network is $K^L$. Due to its exponential nature, this output space will be much larger than the number of samples in the dataset. Figure 1 shows how DDN generates images.

We posit that the contributions of this paper are as follows:

- We introduce a novel generative model, termed Discrete Distribution Networks (DDN), which exhibit a more straightforward and streamlined principle and form.
- For training the DDN, we propose the "Split-and-Prune" optimization algorithm, and a range of practical techniques.
- We conduct preliminary experiments and analysis on the DDN, showcasing its intriguing properties and capabilities, such as zero-shot conditional generation and 1D latent representations.

## 2 Related Work

**Deep Generative Model.** Generative Adversarial Networks (GANs) Radford et al. (2016); Brock et al. (2019) and Variational Autoencoders (VAEs) Kingma & Welling (2014); van den Oord et al. (2018) are two early successful generative models. GANs reduce the divergence between the generated sample distribution and the target distribution through a game between the generator and discriminator. However, regular GANs cannot map samples back to the latent space, thus they cannot reconstruct samples. VAEs encode data into a simple distribution's latent space through an Encoder, and the Decoder is trained to reconstruct the original data from this simple distribution's latent space. Autoregressive models van den Oord et al. (2016) , with their simple principles and methods, model the target distribution by decomposing the target data into conditional probability distributions of each component. They can also compute the exact likelihood of target samples. However, the efficiency of these models is reduced when dealing with image data, which is not suitable to be decomposed into a sequence of components. Normalizing flow Kingma & Dhariwal (2018) is another class of generative models that can compute the likelihood. They use invertible networks to construct a mapping from samples to a noise space, and during the generation stage, they map back from noise to samples. Energy Based Models (EBM) Goyal et al. (2021); Song et al. (2021) with their high-quality and rich generative results, have led to the rise of diffusion models. However, their multi-step iterative generation process requires substantial computational resources. The recent Idempotent Generative Network Shocher et al. (2024) introduces a novel approach by training a neural network to be idempotent, mapping any input to the target distribution effectively.

**Connections to VQ-VAE**. While both VQ-VAE van den Oord et al. (2018) and DDN involve discrete representations, they differ significantly in their approach and capabilities. VQ-VAE enhances the traditional VAE by replacing the continuous latent space with discrete codebooks, thus achieving a more compact representation. VQ-VAE-2 Razavi et al. (2019) further improves this by employing a multi-scale hierarchical structure, thereby enhancing its representational power. However, the discrete representation in VQ-VAE remains two-dimensional, potentially leading to redundant information. Furthermore, VQ-VAE and its successors still rely on an additional prior network for generative modeling in the latent space. Notably, DDN can also serve as this prior model to effectively model the latent space of VQ-VAE. Other distinctions between DDN and VQ-VAE include the absence of an encoder and codebook in DDN, as well as its capacity for Zero-Shot Conditional Generation. VQ-VAEs are known to encounter codebook collapse, a problem that some researchers have addressed by reinitializing unused codes near frequently used ones Williams et al. (2020); Dhariwal et al. (2020). Our Split-and-Prune algorithm shares a similar core idea, albeit with some differences. While the reinitialization method aims to balance code usage to mitigate codebook collapse, our goal is to align the discrete distribution output by our network as closely as possible to the target distribution.

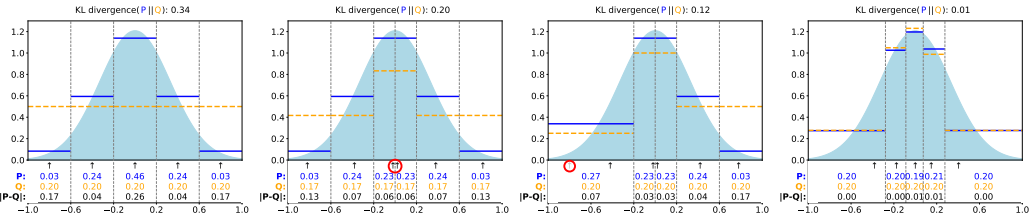
By recursively unfolding the above equations, we can derive the latent variable $\mathbf{k}^*_{1:L}$ and the global objective function $J$.

$$\mathbf{k}^*_{1:L} = [k^*_1, k^*_2, \ldots, k^*_L] = \left[ \underset{k \in \{1,\ldots,K\}}{\operatorname{argmin}} \left\| \mathcal{F}([\mathbf{k}^*_{1:l-1}, k]) - \mathbf{x} \right\|^2 \right]_{l=1}^{L} \tag{3}$$

$$J = \frac{1}{L} \sum_{l=1}^{L} \left\| \mathcal{F}(\mathbf{k}^*_{1:l}) - \mathbf{x} \right\|^2 \tag{4}$$

Here, $\mathcal{F}$ represents the composite function formed from $f_l$, defined as: $\mathcal{F}(\mathbf{k}_{1:l}) = f_l(f_{l-1}(\ldots f_1(\mathbf{x}_0)[k_1]\ldots)[k_{l-1}])[k_l]$. Finally, we average the L2 loss across all layers to obtain the final loss for the entire network.

**How to generation.** When the network performs the generation task, replacing the Guided Sampler with a random choice enables image generation. Given the exponential representational space of $K^L$ sample points and the limited number of samples in the training set, the probability of sampling an image with the same latent space as those in the training set is also exponentially low. For image reconstruction tasks, the process is almost identical to the training process, only substituting the training image with the target image to be reconstructed and omitting the L2 Loss part from the training process. The Final Generated Image in fig. 3a represents the final reconstruction result. The indices of the selected samples along the way form the target image's latent $\mathbf{k}^*_{1:L}$, same as eq. (3). Therefore, the latent $\mathbf{k}^*_{1:L}$ is a sequence of integers with length $L$, which we regard as the hierarchical discrete representation of the target sample. The latent space exhibits a tree structure with $L$ layers and $K$ degrees per node, where each leaf node represents a sample point, and its latent denotes the indices of all nodes along the path to this leaf node, as shown in fig. 1b. In the latent sequence, values placed earlier correspond to higher-level nodes in the tree, controlling the low-frequency information of the output sample, while later values tend to affect high-frequency information.




(a) Initial, K=5 (b) Split, K=5→6 (c) Prune, K=6→5 (d) Final, K=5

Figure 4: **Illustration of the principle behind the Split-and-Prune operation.** For example in (a), the light blue bell-shaped curve represents a one-dimensional target distribution. The 5 "↑" under the x-axis are the initial values from a uniform distribution of 5 output nodes, which divide the entire space into 5 parts using midpoints between adjacent nodes as boundaries (i.e., vertical gray dashed lines). Each part corresponds to the range represented by this output node on the continuous space $x$. Below each node are three values: $P$ stands for the relative frequency of the ground truth falling within this node's range during training; $Q$ refers to the probability mass of this sample (node) in the discrete distribution output by the model during the generation phase, which is generally equal for each sample, i.e., $1/K$. The bottom-most value denotes the difference between $P$ and $Q$. Colorful horizontal line segments represent the average probability density of $P$, $Q$ within corresponding intervals. In (b), the Split operation selects the node with the highest $P$ (circled in red). In (c), the Prune operation selects the node with the smallest $P$ (circled in red). In (d), through the combined effects of loss and Split-and-Prune operations, the distribution of output nodes moves towards final optimization. From the observed results, the KL divergence ($KL(P||Q)$) consistently decreases as the operation progresses, and the yellow line increasingly approximates the light blue target distribution.

preventing the network from generating poor results that are unoptimized. A particular case is when $K = 2$, where the representational space of Taiji-DDN is already sufficiently compact. The use of Chain Dropout in this case tends to result in more blurred generated images. Therefore, we did not employ Chain Dropout when $K = 2$.

## 5 CONCLUSION

In this paper, we have introduced Discrete Distribution Networks, a generative model that approximates the distribution of training data using a multitude of discrete sample points. DDN exhibits unique property: more general zero-shot conditional generation. We also proposed the Split-and-Prune optimization algorithm and several effective techniques for training DDN. Additionally, we showcase the efficacy of DDN and its intriguing properties through experiments.



Figure 6: **Hierarchical Generation Visualization of DDN.** We trained a DDN with output level $L = 3$ and output nodes $K = 8$ per level on MNIST dataset, its latent hierarchical structure is visualized as recursive grids. The large image in the middle is the average of all the generated images. Each sample with a colored border represents an intermediate generation image: a blue border indicates generation by the first layer, and a green border by the second layer. The samples within the surrounding grid of each colored-bordered sample are refined versions generated conditionally based on it (enclosed by the same color frontier). The small samples without colored borders are the final generated images. More detailed visualization of $L = 4$ is presented in the appendix fig. 19.



Figure 7: **Zero-Shot Conditional Generation** on CIFAR-10 Guided by a pretrained classification model without gradient. Each row corresponds to a class in CIFAR-10. Specifically, the first row consists of airplanes, the third row displays birds flying against the blue sky, and the last row presents trucks. Our model successfully generates reasonable images for each class without any conditional training, demonstrating the powerful zero-shot generation capability of our DDN.

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## A Experimental Results on Generative Performance

We present additional experiments to verify the generative and reconstructive capabilities of Discrete Distribution Networks (DDN).

**Conditional DDN for Image-to-Image Tasks.** In the domain of image-to-image tasks, we extend our Discrete Distribution Networks (DDN) to a conditional setting, resulting in the conditional DDN. The image condition is fed directly into the network during training at the beginning of each stage. The condition serves as an informative guide, significantly reducing the task's generative space and, consequently, mitigating the modeling task's complexity. Through this conditional design, the DDN can leverage the abundant information contained in the conditions to generate more accurate and detailed images as shown in fig. 8.

**Verify whether DDN can generate new images.** As depicted in fig. 9, we compare the images that are closest in the training dataset, FFHQ, to those generated by our DDNs. It suggests that our DDNs can synthesize new images that, while not present in the original dataset, still conform to its target distribution.

**Demonstration of Generation and Reconstruction Quality.** Figures 12 to 14 illustrate the results of generation and reconstruction in various ablation experiments using the DDN model.

**Efficacy of Split-and-Prune and Chain Dropout.** A series of experiments were conducted to separately investigate the effectiveness of the Split-and-Prune and Chain Dropout methods. To isolate the impacts of these two algorithms, we simplified the experimental conditions as much as possible, using the MNIST dataset as a base, setting $K = 8$ and $L = 10$, and disabling Learning Residual. The generated image quality under three different settings is displayed in fig. 10. The results demonstrated that the Split-and-Prune method is indispensable, leading to significant improvements in the quality of generated images. Meanwhile, the Chain Dropout method was found to alleviate the poor results observed when the Split-and-Prune method was not implemented.

**Implementation Details of Split-and-Prune.** In algorithm 1, the Split and Prune operations are executed simultaneously. During implementation, the newly split nodes occupy the positions of the pruned nodes in the tensors, ensuring that the total number of network parameters remains constant and the GPU memory usage is fixed. It is important to note that the algorithm must not only operate

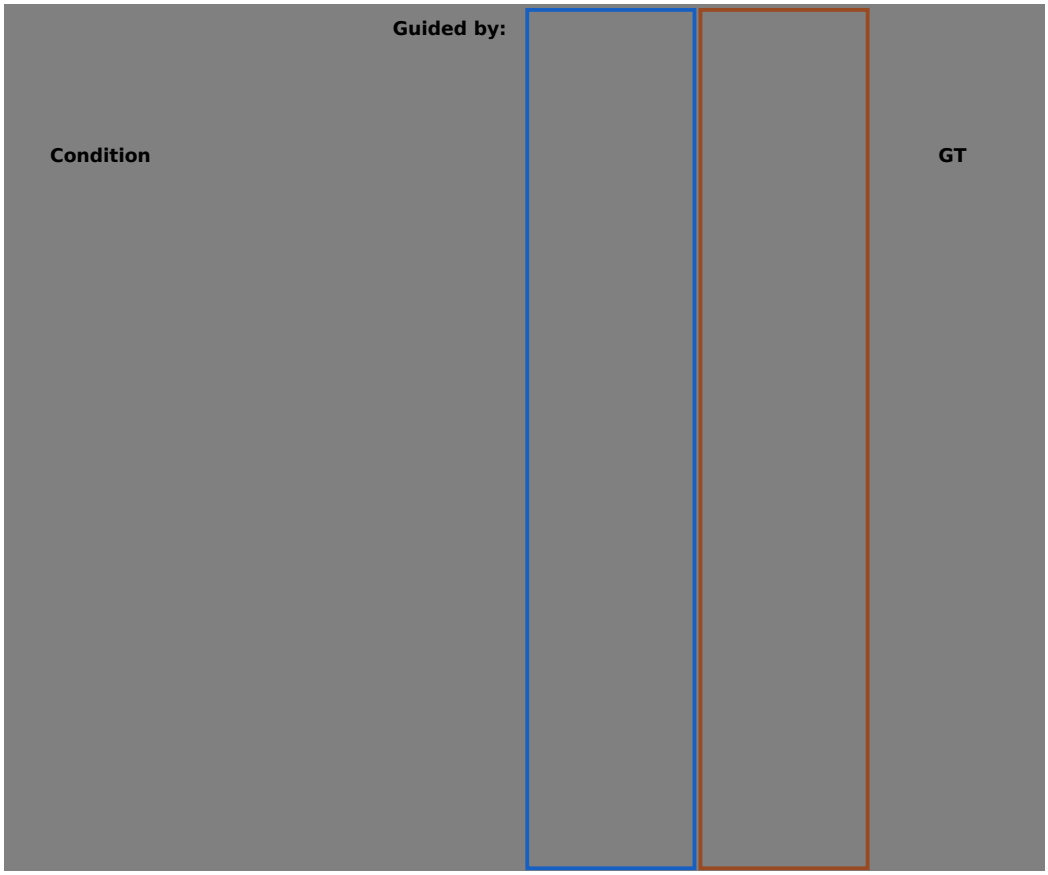



Figure 8: **Conditional DDN performing coloring and edge-to-RGB tasks.** Benefiting from the reduction of the generative space by the condition, DDN is capable of generating high-quality images of $256 \times 256$ resolution. Columns 4 and 5 display combination of conditional generation and ZSCG, the generated results under the guidance of other images, where the produced image strives to adhere to the style of the guided image as closely as possible while ensuring compliance with the condition.